CascadeRoute-VL: Taming the Reasoning Tax in Vision-Language Models via Online Contextual Bandit Routing

Supplementary Material

This supplementary document provides: (i) complete proofs of Lemmas 1–2 (896-d default and 904-d cues-ablation states), Proposition 1 (deterministic horizon-reserve budget feasibility $C_T \le B_{\max} + c_{\max}$), and Theorem 1 (clamped budget-annealed schedule $\alpha_t = \max(\alpha_{\min}, \alpha_0 b_t)$ with explicit finite-time Abbasi-Yadkori constants and OOD Warm-Start prior decay) (Sec. S1); (ii) full 17-VLM profiles, chance-corrected oracle controls, Non-MCQ and strictly dense $\le$ 8B failure rescue tables, upstream MiMo-VL-7B-RL <think> parser repair + int(ANLS<1) truncation audit, unsupervised perceptual diagnostics, and complete 14×17 outcome/token matrices across 30,540 queries (Sec. S2); (iii) Pareto sweeps, 10,000-sample paired bootstrap tests (bounding $p \ge 10^{-4}$), α_0 sensitivity, BwK vs. hard-guardrail budget compliance, regime-scoped calibrated gating, continuous drift/price-shock/onboarding tables, LODO/cross-pool/partial/noisy-feedback generalization tables, and live 4-bit GPU configuration/latency measurements (Sec. S3); (iv) the Counterfactual Off-Policy Evaluation (OPE) validation audit moved from the main text, covering both stationary snapshots and online adaptive trajectories across 10 loggers (Sec. S4); and (v) expanded qualitative case studies and full open-release reproducibility details (Sec. S5).

S1. Complete Theoretical Proofs

S1.1. Proof of Lemma 1 (Bounded Context Space: 896-d Default & 904-d Ablation)

Let our default (896-d Text+Vision) and full (904-d with optional cues) state vectors for query (I_t, Q_t) be

$$s_t^{(896)} = [\, w_q \hat{q}_t \parallel w_v \hat{v}_t \,] \in \mathbb{R}^{896}, \qquad s_t^{(904)} = \left[\, s_t^{(896)} \parallel w_p p_t \,\right] \in \mathbb{R}^{904}$$

where $\hat{q}_t \in \mathbb{R}^{384}$ is L_2-normalized ($\|\hat{q}_t\|_2 = 1$), $\hat{v}_t \in \mathbb{R}^{512}$ is L_2-normalized ($\|\hat{v}_t\|_2 = 1$), and $p_t \in [0,1]^8$ comprises eight bounded unsupervised cues: normalized luminance entropy $H(I_t)/8 \in [0,1]$, Sobel edge gradient fraction in $[0,1]$, logarithmic aspect ratio $\frac{1}{\pi}\arctan(\ln(W/H)) + 0.5 \in (0,1)$, clipped log pixel area $\mathrm{clip}(\log_{10}(WH)/7, 0, 1) \in [0,1]$, bright background fraction $\mathbb{E}[L > 235/255] \in [0,1]$, clipped prompt word ratio $\min(w, 128)/128 \in [0,1]$, symbolic/MCQ cue density in $[0,1]$, and sentence count ratio $\min(s, 16)/16 \in [0,1]$.

Because the three concatenated blocks occupy disjoint coordinate subsets and each entry of p_t lies in $[0,1]$, $\|p_t\|_2^2 \le \sum_{j=1}^{8} 1^2 = 8$. Substituting weights $(w_q, w_v, w_p) = (0.50, 0.55, 0.235)$ yields:

$$\|s_t^{(896)}\|_2^2 = w_q^2 \|\hat{q}_t\|_2^2 + w_v^2 \|\hat{v}_t\|_2^2 = 0.2500 + 0.3025 = 0.5525, \tag{1}$$

$$\|s_t^{(904)}\|_2^2 = \|s_t^{(896)}\|_2^2 + w_p^2 \|p_t\|_2^2 \le 0.5525 + 0.235^2(8) = 0.9943 < 1.0 \tag{2}$$

so $\|s_t^{(896)}\|_2 \equiv \sqrt{0.5525} \approx 0.7433 < 1$ identically on every query, and $\|s_t^{(904)}\|_2 \le \sqrt{0.9943} \approx 0.99715 < 1$. Across all 30,540 encoded real benchmark queries, we empirically verify $\min_t \|s_t^{(904)}\|_2 = 0.7611$, $\mathrm{mean}_t \|s_t^{(904)}\|_2 = 0.8116$, and $\max_t \|s_t^{(904)}\|_2 = 0.8943 < 1.0$. ■

S1.2. Proof of Lemma 2 (Sub-Gaussian Reward Noise)

For every arm $a \in \{1, \dots, K\}$ and round t, accuracy outcome satisfies $y_{t,a} \in \{0,1\}$ and clipped normalized cost satisfies $\tilde{c}_{t,a} = \mathrm{clip}(c_{t,a}/c_{90}, 0, 2.5) \in [0, 2.5]$. With $\gamma \ge 0$, scalar reward $r_{t,a} = y_{t,a} - \gamma \tilde{c}_{t,a}$ is almost surely bounded inside the closed interval $[-2.5\gamma, 1]$ of width $W_\gamma = 1 + 2.5\gamma$. Conditioning on history $\mathcal{F}_{t-1}$ and (s_t, a), the zero-mean noise $\eta_t = r_{t,a} - \mathbb{E}[r_{t,a} \mid s_t, \mathcal{F}_{t-1}]$ is bounded within an interval of width at most $1 + 2.5\gamma$. By Hoeffding's Lemma, for all $\zeta \in \mathbb{R}$:

$$\mathbb{E}\left[\exp(\zeta \eta_t) \mid \mathcal{F}_{t-1}, s_t, a\right] \le \exp\left(\frac{\zeta^2 (1 + 2.5\gamma)^2}{8}\right) = \exp\left(\frac{\zeta^2 \nu^2}{2}\right),$$

establishing ν-sub-Gaussianity with $\nu = \frac{1+2.5\gamma}{2}$. At our balanced setting $\gamma = 0.08$, $\nu = 0.60$. ■

S1.3. Proof of Proposition 1 (Deterministic Horizon-Reserve Budget Feasibility)

Let $B_{\max} > 0$ denote total USD session budget, T the session horizon, $c_{\max} = \max_{t,a} c_{t,a}$ the maximum per-query execution cost, and $c_{\min}^{\mathrm{safe}}$ an upper bound on the cheapest fallback arm $a_{\min}$ (e.g., DeepSeek-VL2-Tiny with mean \$0.0143/1k queries, max per-query cost bound $c_{\min}^{\mathrm{safe}}$). Suppose $B_{\max} \ge T c_{\min}^{\mathrm{safe}}$.

At step $t \in \{1, \dots, T\}$ with prior spend C_{t-1}, define the minimum remaining reserve needed to execute $a_{\min}$ on all future steps $\{t+1, \dots, T\}$ as $R_t :=$

$(T-t)c_{\min}^{\text{safe}}$, and define the step-$t$ feasible arm set $\mathcal{A}_t^{\text{feas}} := \{a \in \mathcal{A} : B_{\max} - (C_{t-1} + \hat{c}_a) \geq R_t\}$, where $\hat{c}_a$ is the per-arm worst-case step cost bound (if $\mathcal{A}_t^{\text{feas}} = \emptyset$, the guardrail deterministically forces $a_t = a_{\min}$). Also set the deferral gate feasibility check to require $B_{\max} - (C_{t-1} + \hat{c}_{a_0} + \hat{c}_{a^*}) \geq R_t$ whenever $b_t > 0.15$.

Consider the last step t^* at which any non-minimum-cost action (with step spend $c_{t^*} > c_{\min}^{\text{safe}}$) is permitted to execute. Because step t^* passed the feasibility check (using realized spend $c_{t^*} \leq \hat{c}_{a_{t^*}}$ on stationary/calibrated schedules, or exceeding $\hat{c}_{a_{t^*}}$ by at most $c_{\max}$ on an adversarial output spike), we have:

$$C_{t^*} = C_{t^*-1} + c_{t^*} \leq B_{\max} - (T-t^*)c_{\min}^{\text{safe}} + c_{\max}. \quad (3)$$

By definition of t^*, every subsequent step $t \in \{t^*+1, \ldots, T\}$ selects $a_{\min}$ whose spend satisfies $c_t \leq c_{\min}^{\text{safe}}$. Summing over those $T - t^*$ trailing queries gives:

$$C_T = C_{t^*} + \sum_{t=t^*+1}^{T} c_t \leq \big(B_{\max} - (T-t^*)c_{\min}^{\text{safe}} + c_{\max}\big) + (T-t^*)c_{\min}^{\text{safe}} = B_{\max} + c_{\max}. \quad (4)$$

When per-step action costs are bounded at selection time by realized/known price caps ($\hat{c}_a \geq c_{t,a}$), the $c_{\max}$ slack vanishes completely ($C_T \leq B_{\max}$), and even with unknown random output lengths, total session overrun is strictly bounded by at most one single query ($c_{\max}$). Table S8 confirms 0/6 hard violations and 0.00% soft overrun across $0.25\times$–$1.25 \times B_{\max}$. ■

S1.4. Proof of Theorem 1 (Clamped-Annealed Disjoint Regret, Numerical Constants, & OOD Prior Decay)

Part A: High-Probability Self-Normalized Confidence Ellipsoid under Clamped Annealing. Let $\mathcal{D}_{\text{off}}$ denote optional warm-start samples scaled by weight $w_0 \in [0,1]$, forming initial Gram matrix $A_a^{(0)} = \lambda I_d + w_0 \sum_{i \in \mathcal{D}_{\text{off}}, a_i = a} s_i s_i^\top$, with $\lambda_{\min}(A_a^{(0)}) \geq \lambda > 0$. During online steps $t = 1, \ldots, T$, let T_a be the number of pulls of arm a ($\sum_{a=1}^K T_a = T$) and $A_{t,a} = A_a^{(0)} + \sum_{\tau < t, a_\tau = a} s_\tau s_\tau^\top$. By Theorem 2 of Abbasi-Yadkori et al. [1] applied with Lemma 1 ($\|s_t\|_2 \leq 1$) and Lemma 2 (ν-sub-Gaussian noise) under stationary rewards ($\mathbb{E}[r_{t,a}|s_t] = \theta_a^\top s_t$, $\|\theta_a\|_2 \leq S$), using the trace-determinant inequality $\ln(\det(A_{t,a})/\det(A_a^{(0)})) \leq d\ln(1 + T/(d\lambda))$, with probability at least $1-\delta$, simultaneously for all $t \geq 1$ and $a \in \{1, \ldots, K\}$:

$$\|\hat{\theta}_{t,a} - \theta_a\|_{A_{t,a}} \leq \beta_T(\delta/K) := \nu\sqrt{2d\ln\left(\left(1 + \frac{T}{d\lambda}\right)\frac{K}{\delta}\right)} + \sqrt{\lambda}S, \quad (5)$$

Consider the clamped budget-annealed schedule $\alpha_t = \max(\alpha_{\min}, \alpha_0 b_t)$ where $b_t \in [0,1]$ and $\alpha_{\min} \geq \beta_T(\delta/K)$. Define $\bar{\alpha} := \max(\alpha_{\min}, \alpha_0) \geq \alpha_t \geq \alpha_{\min} \geq \beta_T(\delta/K)$ for all $t \in \{1, \ldots, T\}$. Because $\alpha_t \geq \|\hat{\theta}_{t,a} - \theta_a\|_{A_{t,a}}$ at every round, Cauchy-Schwarz in the $A_{t,a}^{-1}$-induced metric gives:

$$\left|(\hat{\theta}_{t,a} - \theta_a)^\top s_t\right| \leq \|\hat{\theta}_{t,a} - \theta_a\|_{A_{t,a}} \|s_t\|_{A_{t,a}^{-1}} \leq \alpha_t \sqrt{s_t^\top A_{t,a}^{-1} s_t}. \quad (6)$$

Notice why fixing $\alpha_t = \max(\alpha_{\min}, \alpha_0 b_t)$ matters: without clamping by $\alpha_{\min} \geq \beta_T(\delta/K)$, optimism fails once $\alpha_0 b_t$ decays below the confidence width. With the clamp active, the UCB rule $a_t = \arg\max_a (\hat{\theta}_{t,a}^\top s_t + \alpha_t \|s_t\|_{A_{t,a}^{-1}})$ against hindsight optimal arm $a_t^* = \arg\max_a \theta_a^\top s_t$ satisfies:

$$\begin{aligned}
\text{reg}_t &= \theta_{a_t^*}^\top s_t - \theta_{a_t}^\top s_t \\
&\leq \hat{\theta}_{t,a_t^*}^\top s_t + \alpha_t \|s_t\|_{A_{t,a_t^*}^{-1}} - \theta_{a_t}^\top s_t \\
&\leq \hat{\theta}_{t,a_t}^\top s_t + \alpha_t \|s_t\|_{A_{t,a_t}^{-1}} - \theta_{a_t}^\top s_t \\
&\leq 2\alpha_t \sqrt{s_t^\top A_{t,a_t}^{-1} s_t} \leq 2\bar{\alpha}\sqrt{s_t^\top A_{t,a_t}^{-1} s_t}. \quad (7)
\end{aligned}$$

Since rewards lie in $[-2.5\gamma, 1]$, clipping per-step regret above by $W_\gamma = 1 + 2.5\gamma$ gives $\text{reg}_t \leq 2\bar{\alpha}\min(1, \|s_t\|_{A_{t,a_t}^{-1}})$ (whenever $\bar{\alpha} \geq W_\gamma/2$). Grouping rounds by arm $a \in \{1, \ldots, K\}$ with counts T_a ($\sum_a T_a = T$) and applying Cauchy-Schwarz first per arm, then across the K arms via $\sum_{a=1}^K \sqrt{T_a} \leq \sqrt{KT}$:

$$\begin{aligned}
R_T = \sum_{t=1}^T \text{reg}_t &\leq 2\bar{\alpha} \sum_{a=1}^K \sum_{t:\, a_t = a} \min\left(1, \|s_t\|_{A_{t,a}^{-1}}\right) \\
&\leq 2\bar{\alpha} \sum_{a=1}^K \sqrt{T_a \sum_{t:\, a_t = a} \min\left(1, \|s_t\|_{A_{t,a}^{-1}}^2\right)}. \quad (8)
\end{aligned}$$

By the Determinant-Trace Elliptical Potential Lemma [1] with $\det(A_{t,a}) = \det(A_a^{(0)}) \prod_{\tau: a_\tau = a}(1 + \|s_\tau\|_{A_{\tau,a}^{-1}}^2)$ and Hadamard's inequality ($\text{tr}(A_{T+1,a}) \leq d\lambda_{\max}^{(0)} + T_a$), we obtain $\sum_{t:a_t=a} \min(1, \|s_t\|_{A_{t,a}^{-1}}^2) \leq 2\ln(\det(A_{T+1,a})/\det(A_a^{(0)})) \leq 2d\ln(1 + \frac{T_a}{d\lambda})$. Substituting back gives:

$$R_T \leq 2\max(\alpha_{\min}, \alpha_0)\sqrt{2dKT\ln\left(1 + \frac{T}{d\lambda}\right)} + \mathcal{O}(K) = \tilde{\mathcal{O}}(d\sqrt{KT}). \quad (9)$$

Explicit Numerical Constants & Practical Regime. Substituting our worst-case operational constants (d=904, K=17, T=6108, λ=4.0, ν=0.60 at γ=0.08, empirical Dev-estimated $\|\hat{\theta}_a\|_2 \leq S$=3.329, δ=0.05)

yields:

$$\alpha_{\min}^{\text{theory}} = 0.60\sqrt{2(904)\ln\left(\left(1+\frac{6108}{904\cdot 4.0}\right)\frac{17}{0.05}\right)}+\sqrt{4.0}(3.329)\approx 36.01+6.66=42.67,$$

with worst-case bound multiplier $2(42.67)\sqrt{2(904)(17)(6108)\ln(2.689)} \approx 1.16\times 10^6$ (trivial vs. horizon T=6108 under d=904 worst-case ellipsoid looseness; for d=896, $\alpha_{\min}^{\text{theory}} = 42.51$). As shown empirically in Table S8, setting α_0=20 approaches pure uniform exploration and degrades OOD Test accuracy to 77.15%, whereas Dev-tuned scaling (α_0=0.10, $\alpha_{\min}$=0.01) captures the uncertainty bonus inside the plateau (78.49%–78.59% i.i.d.; 80.85%–80.96% OOD). Thus, Eq. (7) provides the principled clamped regret condition for theoretical soundness, while empirical deployment uses Dev-tuned (α_0=0.10, $\alpha_{\min}$=0.01).

Part B: Why Offline Routers Fail OOD While Warm-Start Prior Decays. Consider an OOD target distribution where the true parameter vector shifts from offline source θ_a^{src} to θ_a^{ood} with suboptimality gap $\Delta_{\text{shift}} > 0$ on a non-negligible fraction of OOD queries. Any frozen offline supervised classifier (Offline-Linear, ARMS, or MLP-Router) never updates weights during OOD inference, incurring strictly linear OOD regret $R_T^{\text{offline}} = \Omega(\Delta_{\text{shift}} T)$. In contrast, consider CascadeRoute-VL solving Ridge with offline Gram matrix scaled by $w_0 \in (0,1)$ ($A_a^{(0)} = \lambda I_d + w_0 X_{\text{src},a}^\top X_{\text{src},a}$). After observing $n_{t,a}$ OOD pulls of arm a via online Sherman-Morrison updates:

$$\hat{\theta}_{t,a} - \theta_a^{\text{ood}} = A_{t,a}^{-1}\Big[w_0 X_{\text{src},a}^\top X_{\text{src},a}(\theta_a^{\text{src}} - \theta_a^{\text{ood}}) - \lambda\theta_a^{\text{ood}} + \sum_{\tau\le t,\, a_\tau = a} \eta_\tau s_\tau \Big]. \quad (10)$$

Since empirical OOD covariance grows linearly along active OOD directions ($\lambda_{\min}(A_{t,a}) = \Omega(\lambda + n_{t,a})$ on the spanned subspace), the stale source bias decays inversely with online sample count as $\mathcal{O}\left(\frac{w_0\|\theta_a^{\text{src}} - \theta_a^{\text{ood}}\|}{\lambda + n_{t,a}}\right)$. Using moderate unified prior weight (w_0=0.20) delivers low initial variance while guaranteeing rapid Bayesian prior forgetting under OOD shift. ■

S2. Complete 30,540-Query Benchmark & Perceptual Data

Tables S1, S2, S4, and S5 document the complete empirical profiles reconstructed from all 238 VLMEvalKit inference files across 14 benchmarks and 17 VLMs (30,540 × 17 = 519,180 query-model pairs, > 99.9% exact cell coverage).

S3. Extended Results & Statistical Significance

Tables S6, S7, and S12 report full Pareto sweep [0.0, 0.9], Bonferroni-corrected hypothesis tests against all primary baselines, and CascadeRoute-VL's learned zero-label routing distribution across all four domains and 17 VLM arms.

S4. Counterfactual Off-Policy Evaluation (OPE) Audit

Because our primary release provides the complete VL-RouterBench counterfactual outcome tensor $Y \in \{0,1\}^{30,540\times 17}$ (together with the <think>-repaired matrix), direct policy accuracy evaluation is exact. As an independent sanity check for production scenarios where only logged single-arm traces exist, Table S11 reports Inverse Propensity Scoring (IPS), Self-Normalized IPS (SNIPS), and Doubly Robust (DR) [2, 3] 95% confidence intervals across 10 uniformized behavior loggers (ϵ=0.75 exploration mix with a Ridge-fitted behavior propensity distribution and 10k paired bootstrap intervals).

Crucially, Table S11 evaluates both the frozen Warm-Start Snapshot (0.7878 true, 0.7942 DR) and the sequential Online Adaptive Trajectory (0.7845 true, 0.7894 DR, [0.7496, 0.8255], Effective Sample Size ESS=397.7, achieving 10/10 empirical CI coverage across 10 independent behavior seeds). We note explicitly that under skewed loggers where greedy target arms receive tiny propensity weights $p_{\log}(a|x) < 0.01$, SNIPS and DR CIs suffer high effective-sample variance and can exclude true static accuracy (as observed with Qwen2.5-VL-72B before uniformized logging support), reinforcing why complete counterfactual matrices remain essential alongside logged OPE checks.

S5. Qualitative Reasoning Tax Gallery & Reproducibility

Figure S1 displays four authentic benchmark queries extracted verbatim from VL-RouterBench transcripts illustrating both failure modes CascadeRoute-VL resolves:

- OCRBench #314 (Yellow Cab): Ground truth "Y809". Qianfan-8B reads "The cab number is Y809." (7 tok, right); Qwen2.5-72B (17 tok), Kimi-Think (39 tok), and MiMo-VL-RL (136 tok) confuse the fleet phone "426-6262" or fabricate "7007".
- OCRBench #979 (LaTeX Inequality): Ground truth "$m \neq \sqrt{2}$". Qianfan-8B outputs "m \neq \sqrt { 2 }" (8 tok, right); Qwen2.5-72B, Kimi-Think, and DS-Tiny misread ≠ as +, while MiMo-VL-RL emits 189 speculative <think> tokens ("$M + \sqrt{x}$").

Table S1. Complete Profiles of All 17 VLMs Across N=30,540 Queries. Token counts measured using cl100k_base (+256 image tokens/query); pricing follows VL-RouterBench schedules; latencies computed via class TTFT + decode throughput profiles OLS-calibrated to RTX 4050 measurements. †MiMo-VL-7B-RL reports 35.83% raw in Y.npz due to unstripped <think> OCR extraction failures, rising to 77.76% global after <think> repair.

Model	Params / Type	Acc (%)	Avg In Tok	Avg Out Tok	Think Tag (%)	In $/1M	Out $/1M	Cost/1k ($)	Lat (s)
Qwen2.5-VL-72B-Instruct	72B (Local)	78.21	278.3	33.3	0.0	0.80	1.20	$0.2626	2.032
InternVL2.5-78B	78B (Local)	76.63	278.3	15.8	0.0	1.00	1.50	$0.3020	1.348
Qwen2.5-VL-32B-Instruct	32B (Local)	74.50	278.3	104.4	0.0	0.40	0.60	$0.1740	3.410
Kimi-VL-A3B-Thinking-2506	3B-MoE (16B Tot)	73.80	278.3	59.4	0.0	0.20	0.25	$0.0705	0.765
GeminiFlash2.5	API (Proprietary)	72.33	278.3	118.1	0.0	0.30	2.40	$0.3670	1.394
DeepSeek-VL2	27B (Local)	70.91	278.3	18.7	0.0	0.35	0.50	$0.1068	0.726
Qianfan-VL-8B	8B (Local)	70.61	278.3	4.3	0.0	0.18	0.25	$0.0512	0.231
DeepSeek-VL2-Tiny	1.5B (Local)	63.23	278.3	7.9	0.0	0.05	0.05	$0.0143	0.148
Gemma3-27B	27B (Local)	55.11	278.3	75.6	0.0	0.35	0.50	$0.1352	2.290
GPT-4o	API (Proprietary)	53.01	278.3	29.2	0.0	2.50	10.00	$0.9882	0.967
Phi-3.5-Vision	7B (Local)	51.62	278.3	6.0	0.0	0.10	0.10	$0.0284	0.197
SmolVLM2	2B (Local)	51.55	278.3	6.8	0.0	0.06	0.06	$0.0171	0.154
Pixtral-12B	12B (Local)	44.66	278.3	39.3	0.0	0.25	0.35	$0.0833	0.935
LLaVA-NeXT-Vicuna-7B	7B (Local)	44.54	278.3	20.2	0.0	0.20	0.20	$0.0597	0.429
MiMo-VL-7B-RL	7B (Local)	35.83 / 77.76†	278.3	237.4	100.0	0.20	0.30	$0.1269	3.467
Janus-Pro-7B	7B (Local)	31.37	278.3	11.0	0.0	0.18	0.25	$0.0528	0.306
Janus-Pro-1B	1B (Local)	28.84	278.3	23.3	0.0	0.05	0.05	$0.0151	0.241

Table S2. Per-Dataset Unsupervised Perceptual Cues & Rescue Breakdown Across 14 Benchmarks (N=30,540). Strictly dense ≤ 8B models resolve 63.2% (4,209/6,655) of Qwen2.5-VL-72B failures (68.5% when adding 3B-active/16B-total MoE Kimi-VL-A3B) using zero domain labels.

Dataset	Domain	N	Ent (b)	Sobel	WhiteBg	Words	MathCue	Oracle	72B	Best ≤8B / 3B-MoE (Acc)	72B Fails Rescued
AI2D_TEST	STEM	3,088	3.34	0.162	0.710	9.6	0.005	99.68	88.96	Qianfan-8B (85.04)	90.9%
MathVerse_MINI	STEM	3,940	2.00	0.117	0.887	39.8	0.359	90.10	51.50	MiMo-7B-RL (66.50)	67.5%
MathVision_MINI	STEM	304	2.39	0.133	0.835	61.6	0.292	92.11	38.82	Kimi-3B-MoE (47.04)	69.4%
MathVista_MINI	STEM	1,000	3.72	0.136	0.600	41.4	0.268	96.10	73.90	Kimi-3B-MoE (79.60)	75.9%
MMBench_DEV_EN	General	1,292	6.28	0.157	0.162	8.9	0.002	97.91	88.24	Qianfan-8B (84.67)	72.4%
MMMU_DEV_VAL	General	1,050	3.91	0.175	0.578	33.0	0.269	97.33	67.90	Kimi-3B-MoE (60.48)	83.1%
MMStar	General	1,500	5.11	0.152	0.372	16.8	0.029	96.73	71.40	MiMo-7B-RL (70.87)	81.8%
RealWorldQA	General	765	6.94	0.142	0.005	8.4	0.016	99.35	75.42	MiMo-7B-RL (72.55)	87.2%
HallusionBench	Halluc.	951	3.93	0.169	0.522	16.2	0.123	98.74	72.87	Kimi-3B-MoE (72.98)	92.6%
ChartQA_TEST	OCR	2,500	3.15	0.095	0.761	11.4	0.236	97.12	87.20	Qianfan-8B (86.48)	70.6%
DocVQA_VAL	OCR	5,349	3.69	0.177	0.485	8.5	0.065	98.47	91.21	Qianfan-8B (87.42)	71.7%
InfoVQA_VAL	OCR	2,801	5.54	0.281	0.309	11.1	0.110	96.14	82.86	Qianfan-8B (67.37)	53.1%
OCRBench	OCR	1,000	5.32	0.218	0.296	10.3	0.044	98.50	88.70	Qianfan-8B (85.70)	72.6%
TextVQA_VAL	OCR	5,000	7.10	0.163	0.054	7.1	0.009	89.44	75.82	Qianfan-8B (73.78)	48.5%

- MathVista #897 & #874 (Trigonometry & Step-Function): Compact perception arms guess wrong in 1–5 tokens and Qwen2.5-72B misidentifies cos vs. sin (197 tok, wrong) and step-interval endpoints (157 tok, wrong), while Kimi-VL-A3B-Thinking derives exact answers ("40" in 178 tok; "2" in 269 tok).

S5.1. Reproducibility & Open-Release Package

All experiments execute in Python 3.13.5 + PyTorch 2.8.0 CUDA (RTX 4050 6.44 GB GPU) with offline local HuggingFace/TorchHub weights (all-MiniLM-L6-v2, DINOv2-ViT-S/14, ResNet-50, plus 4-bit Qwen3-VL-2B-Instruct and -Thinking). Feature weights (w_q, w_v, w_p)=$(0.50, 0.55, 0.235)$ are fixed analytically by Lemma 1, with 896-d Text+Vision serving as default after the −2.41% OOD cues ablation. Bandit hyperparameters (α_0=0.10, $\alpha_{\min}$=0.01, λ=4.0, unified w_0=0.20, discounted β=0.996, Dev-calibrated deferral threshold τ_{cal}=0.04 replacing earlier heuristic gate constants) were tuned strictly on the Dev split (N=3,054), and all Test (N=6,108) and OOD (N=17,601) streaming metrics report 5-seed mean ± std plus 10,000-resample paired bootstrap CIs. We release our standardized raw/repaired outcome tensors (Y.npz, Y_repaired.npz), all 238 parsed VLMEvalKit inference transcripts metadata, unit-test suite (pytest tests/ -v), and reproducible experiment scripts (scripts/exp1_*.py–exp4_*.py).

Table S3. Claim 1 Controls: Split Consistency, Independent-Chance Oracle Check, Dense ≤ 8B vs. Non-MCQ Rescues, & Upstream Parser Repair. Raw 17-arm oracle values are lower than independent-chance controls (≈ 100%) due to cross-model error correlation; genuine complementarity is verified via Non-MCQ rescues and MiMo-VL <think> parsing recovery.

Split / Control Check	N	72B (%)	Real Oracle	Strat. Chance	Gap vs Best	Failure Rescue Subset	Matrix	Dense ≤8B	+3B-MoE	Any Other
Full VL-RouterBench (Raw)	30,540	78.21	95.45	99.99	+17.24	All 14 Benchmarks (6,655 fails)	Raw Y	63.2%	68.5%	79.1%
Held-Out Test (Raw)	6,108	77.29	95.17	100.00	+17.88	Non-MCQ Only (5,138 fails)	Raw Y	57.6%	63.6%	75.4%
Non-MCQ Subset (Raw)	22,659	77.32	94.43	99.99	+17.10	MCQ Only (1,517 fails)	Raw Y	82.2%	85.0%	91.7%
In-Dist Test (Gen+STEM)	2,523	69.24	95.48	99.99	+22.35	All 14 Benchmarks (6,655 fails)	MiMo-Repaired	67.6%	71.9%	81.4%
MiMo-Repaired Exact (Full)	30,540	78.21	95.95	99.99	+17.74	Non-MCQ Only (5,138 fails)	MiMo-Repaired	63.3%	68.0%	78.4%
All-17 Repaired (ANLS ≥ 0.5)	30,540	81.39	97.23	99.99	+15.83	All 14 Benchmarks (5,682 fails)	All-17 ≥ 0.5	70.6%	75.0%	85.1%

Upstream Extraction Repair Audit: MiMo-VL-7B-RL Raw 0.0% → ChartQA: 84.72%, DocVQA: 88.67% exact / 95.51% (≥ 0.5, 94.24% cont.), InfoVQA: 79.76% / 85.18% (83.70% cont.), TextVQA: 74.16% / 78.60% (35.83% → 77.76% global; 0.0–0.3% unclosed). Upstream int(ANLS<1) truncation: SmolVLM2 InfoVQA 0.0% → 49.66% (≥ 0.5, 37.75% ANLS); Janus-Pro-1B/7B DocVQA 0.60%/1.08% → 46.91%/52.91% (≥ 0.5).

Table S4. Full 14 × 17 Benchmark Accuracy Matrix (%). Abbreviations: GPT4o, Gem2.5 (GeminiFlash2.5), Gem27 (Gemma3-27B), Int78 (InternVL2.5-78B), J1B/J7B (Janus-Pro-1B/7B), Kimi (Kimi-VL-A3B-Thinking), MiMo (MiMo-VL-7B-RL), Phi3.5, Pix12 (Pixtral-12B), Qian8 (Qianfan-VL-8B), Q32/Q72 (Qwen2.5-VL-32B/72B), Smol2 (SmolVLM2), DS27/DST (DeepSeek-VL2/Tiny), LLaVA7.

Dataset	GPT4o	Gem2.5	Gem27	Int78	J1B	J7B	Kimi	MiMo	Phi3.5	Pix12	Qian8	Q32	Q72	Smol2	DS27	DST	LLaVA7	Oracle
AI2D	83.2	88.6	81.4	88.9	66.9	68.0	82.6	83.3	77.6	44.7	85.0	85.6	89.0	69.6	82.8	74.6	66.8	99.7
ChartQA	36.7	38.7	30.0	86.0	23.2	25.7	73.0	0.0	71.2	63.3	86.5	70.2	87.2	68.6	85.8	81.1	54.0	97.1
DocVQA	38.5	85.9	56.1	89.1	0.6	1.1	87.1	0.0	54.1	59.0	87.4	87.5	91.2	63.9	85.1	80.1	50.2	98.5
Hallusion	69.1	76.6	64.0	72.3	51.1	60.3	73.0	72.3	59.3	46.4	69.3	72.1	72.9	60.2	64.4	61.6	47.2	98.7
InfoVQA	28.6	77.4	28.4	77.3	0.0	0.0	66.3	0.0	25.1	18.5	67.4	76.4	82.9	0.0	67.5	55.6	20.0	96.1
MMBench	82.2	89.6	80.4	89.2	58.7	64.5	84.0	84.1	71.8	60.8	84.7	86.5	88.2	67.4	82.5	70.9	65.6	97.9
MMMU	68.1	77.9	62.8	66.5	37.1	42.0	60.5	59.5	44.3	32.5	48.4	69.9	67.9	41.6	54.2	42.0	38.5	97.3
MMStar	61.1	77.0	60.6	69.5	43.1	46.1	69.9	70.9	47.2	41.4	63.3	69.7	71.4	45.9	62.2	49.5	37.8	96.7
MathVerse	39.1	50.8	41.4	48.5	17.0	20.9	61.0	66.5	22.9	15.8	31.1	54.5	51.5	22.5	37.0	23.9	13.4	90.1
MathVision	32.2	59.2	33.2	30.3	16.8	17.1	47.0	35.9	16.8	15.8	18.1	36.5	38.8	17.4	22.0	16.1	14.5	92.1
MathVista	57.7	80.2	60.6	70.6	35.7	43.0	79.6	77.8	43.7	41.8	64.0	73.3	73.9	51.5	63.7	53.8	31.5	96.1
OCRBench	72.0	85.6	75.1	85.0	49.0	59.3	82.8	84.6	60.3	53.0	85.7	85.4	88.7	72.9	81.3	80.3	50.4	98.5
RealWorldQA	68.5	76.7	62.5	78.4	54.8	42.1	70.9	72.6	53.9	63.8	70.5	69.8	75.4	57.5	69.9	64.7	58.6	99.4
TextVQA	60.7	66.7	59.9	76.8	37.3	40.4	69.5	0.0	58.5	54.1	73.8	71.4	75.8	65.4	76.5	72.5	56.8	89.4

Table S5. Full 14 × 17 Mean Generated Output Token Matrix (cl100k_base). Notice how MiMo-VL-7B-RL emits up to 1,401.7 output tokens on MathVision yet collapses to 4.8–55.7 broken tokens on OCR datasets, whereas Qianfan-VL-8B answers OCR in 1.2–4.6 tokens.

Dataset	GPT4o	Gem2.5	Gem27	Int78	J1B	J7B	Kimi	MiMo	Phi3.5	Pix12	Qian8	Q32	Q72	Smol2	DS27	DST	LLaVA7
AI2D	13.7	233.2	97.9	1.0	1.0	1.4	113.5	273.7	1.1	33.8	1.0	131.7	3.7	1.0	19.1	1.0	1.0
ChartQA	1.5	4.8	2.2	1.2	1.2	1.2	14.7	44.8	1.2	1.2	1.2	1.5	1.2	1.2	1.2	1.2	1.2
DocVQA	2.1	4.2	2.6	2.8	3.0	2.5	5.0	55.7	2.9	2.2	2.7	2.9	2.9	2.8	2.8	2.7	2.8
Hallusion	50.8	75.2	77.9	1.0	1.5	1.0	70.6	335.4	18.9	42.7	24.1	164.2	58.6	1.0	23.6	9.6	64.2
InfoVQA	1.6	2.4	2.0	1.9	2.6	1.7	22.1	5.5	1.6	1.2	1.9	2.9	2.0	1.8	1.9	2.6	2.0
MMBench	12.0	169.2	104.1	1.0	1.0	1.8	67.4	269.1	1.1	32.1	1.0	63.6	7.1	1.0	2.5	1.2	1.0
MMMU	173.9	520.1	321.9	15.6	3.9	28.5	2.3	717.3	12.0	113.8	5.1	419.0	32.0	6.8	20.2	4.8	7.6
MMStar	58.3	261.3	152.3	1.0	1.1	3.4	1.2	377.5	1.3	60.2	1.0	152.2	4.9	1.0	9.7	1.1	1.3
MathVerse	56.2	249.9	198.2	40.0	160.1	59.9	209.7	702.1	20.6	147.6	2.1	347.7	144.6	37.8	102.6	43.6	74.3
MathVision	355.0	846.6	554.9	367.5	23.5	5.0	346.3	1401.7	33.4	216.7	135.9	521.4	375.0	4.2	5.0	5.0	195.2
MathVista	135.8	264.8	176.0	153.6	17.4	7.1	142.5	491.1	4.1	107.0	9.1	245.2	156.8	4.8	5.0	5.2	113.6
OCRBench	17.7	19.5	32.2	1.9	9.3	14.7	46.0	125.8	18.9	17.3	4.6	41.7	19.9	4.2	7.0	6.2	41.7
RealWorldQA	7.5	106.6	52.2	1.0	1.0	1.1	19.3	209.3	1.1	9.8	1.1	27.1	6.2	4.2	1.0	1.0	1.0
TextVQA	1.7	2.2	1.9	1.8	1.7	2.2	8.8	4.8	1.7	1.4	1.7	2.2	1.9	1.7	1.8	1.8	1.7

Table S6. Full Pareto Sweep Data Corresponding to Main Paper Figure 3 (Test N=6,108). CascadeRoute-VL maintains smooth monotonic accuracy-cost trade-offs across the entire cost spectrum.

Method (γ)	Acc (%)	Cost/1k	Tok	Lat (s)	Method (γ/τ)	Acc (%)	Cost/1k	Tok	Lat (s)
CascadeRoute-VL (0.00)	80.42	$0.2840	395.7	1.959	KNN-Router (0.00)	80.29	$0.2812	369.6	1.667
CascadeRoute-VL (0.03)	79.99	$0.2316	388.4	1.840	KNN-Router (0.08)	78.31	$0.1306	361.8	1.444
CascadeRoute-VL (0.08)	78.49	$0.1585	368.6	1.516	KNN-Router (0.30)	75.95	$0.0753	344.4	1.091
CascadeRoute-VL (0.15)	77.44	$0.1115	359.2	1.180	KNN-Router (0.90)	73.05	$0.0431	320.6	0.653
CascadeRoute-VL (0.30)	75.93	$0.0879	360.4	0.831	ARMS-Router (0.00)	80.14	$0.3063	371.0	1.411
CascadeRoute-VL (0.55)	75.33	$0.0844	377.4	0.699	ARMS-Router (0.08)	78.80	$0.2190	369.0	1.306
CascadeRoute-VL (0.90)	74.36	$0.0818	410.1	0.672	ARMS-Router (0.90)	72.22	$0.0448	321.3	0.589
MLP-Router (0.00)	81.01	$0.2834	406.3	2.068	Offline-Linear (0.00)	79.68	$0.2718	386.0	1.817
MLP-Router (0.03)	80.47	$0.2399	406.0	2.006	Offline-Linear (0.08)	78.21	$0.1777	384.3	1.721
MLP-Router (0.08)	79.85	$0.1854	400.4	1.844	Offline-Linear (0.90)	71.37	$0.0470	328.3	0.715
MLP-Router (0.15)	78.65	$0.1412	390.0	1.656	FrugalGPT (τ=0.55)	69.38	$0.0641	472.9	0.777
MLP-Router (0.30)	76.31	$0.0883	365.9	1.289	FrugalGPT (τ=0.65)	71.12	$0.1058	579.2	1.184
MLP-Router (0.55)	74.79	$0.0588	345.4	0.984	FrugalGPT (τ=0.72)	73.23	$0.1704	725.8	1.719
MLP-Router (0.90)	72.69	$0.0470	330.9	0.761	FrugalGPT (τ=0.88)	76.74	$0.3515	1121.1	2.906

Table S7. 10,000-Resample Paired Bootstrap Confidence Intervals Over Test Queries (N=6,108, Seed 42). Permutation p-values are strictly resolution-bounded ($p \geq 10^{-4}$ for 10,000 resamples), replacing extreme parametric normality assumptions.

Comparison (CascadeRoute-VL vs.)	Acc Δ (%)	Acc 95% Boot CI	Boot p	USD Saved (/1k)	Cost 95% Boot CI	Cost p
Qwen2.5-VL-72B	+1.20	[+0.23, +2.14]	1.52×10^{-2}	+$0.1021	[+0.0967, +0.1073]	1.0×10^{-4}
Greedy Warm-Start (α=0)	+0.20	[+0.00, +0.39]	6.74×10^{-2}	-$0.0010	[−0.0020, −0.0001]	3.01×10^{-2}
Per-Domain Best Static	+0.38	[−0.51, +1.24]	4.19×10^{-1}	+$0.0283	[+0.0230, +0.0335]	1.0×10^{-4}
Offline-Linear	+0.28	[−0.52, +1.11]	5.18×10^{-1}	+$0.0192	[+0.0127, +0.0261]	1.0×10^{-4}
KNN-Reward (cost-aware)	+0.29	[−0.67, +1.26]	5.60×10^{-1}	-$0.0451	[−0.0515, −0.0386]	1.0×10^{-4}
ARMS-Router	-0.31	[−1.15, +0.54]	4.82×10^{-1}	+$0.0605	[+0.0519, +0.0694]	1.0×10^{-4}
MLP-Router (BCE, i.i.d. static)	-1.33	[−2.10, −0.59]	1.0×10^{-3}	+$0.0273	[+0.0198, +0.0352]	1.0×10^{-4}
MLP-Reward (MSE, i.i.d. static)	-1.16	[−1.90, −0.43]	2.3×10^{-3}	+$0.0206	[+0.0144, +0.0273]	1.0×10^{-4}
NeuralUCB (Warm+Online)	+3.39	[+2.41, +4.40]	1.0×10^{-4}	+$0.0499	[+0.0413, +0.0588]	1.0×10^{-4}
NeuralTS (Warm+Online)	+2.78	[+1.80, +3.78]	1.0×10^{-4}	+$0.0395	[+0.0307, +0.0488]	1.0×10^{-4}

Table S8. Deterministic Horizon-Reserve Guardrail (Theorem 1) vs. BwK Primal-Dual & α_0 Sensitivity (Test N=6,108). Hard-Guarded CascadeRoute attains 0/6 violations across all budgets and beats BwK in accuracy at every tier.

Budget ($B_{\max}$)	Method	Spend ($)	Ratio	Viol?	Soft Over%	Acc (%)	α_0	i.i.d. Acc	i.i.d. $/1k	OOD Acc	OOD $/1k
0.25× ($0.2443)	Hard-Guarded (Thm 1)	0.2443	1.000	No	0.00	65.32	0.00	78.29	0.1574	81.02	0.1305
0.25× ($0.2443)	BwK Primal-Dual	0.3182	1.302	Yes	30.23	65.28	0.05	78.31	0.1577	80.85	0.1329
0.40× ($0.3909)	Hard-Guarded (Thm 1)	0.3909	1.000	No	0.00	68.37	0.10	78.49	0.1585	80.96	0.1267
0.40× ($0.3909)	BwK Primal-Dual	0.4562	1.167	Yes	16.70	67.21	0.25	78.45	0.1578	80.08	0.1125
0.60× ($0.5864)	Hard-Guarded (Thm 1)	0.5864	1.000	No	0.00	71.69	0.50	78.59	0.1596	79.10	0.0904
0.60× ($0.5864)	BwK Primal-Dual	0.6408	1.093	Yes	9.29	69.52	1.00	78.45	0.1585	80.38	0.1167
0.80× ($0.7818)	Hard-Guarded (Thm 1)	0.7818	1.000	No	0.00	75.95	2.50	78.29	0.1593	80.96	0.1358
0.80× ($0.7818)	BwK Primal-Dual	0.8240	1.054	Yes	5.39	71.74	5.00	78.29	0.1597	80.40	0.1218
1.00× ($0.9773)	Hard-Guarded (Thm 1)	0.9675	0.990	No	0.00	78.45	10.0	78.37	0.1606	78.94	0.1325
1.00× ($0.9773)	BwK Primal-Dual	1.0089	1.032	Yes	3.24	74.31	20.0	78.09	0.1634	77.15	0.1439
1.25× ($1.2216)	Hard-Guarded (Thm 1)	0.9637	0.789	No	0.00	78.52	$\alpha_{\min}^{\text{theory}}$=42.67 ($d$=904, T=6108, λ=4, S=3.33)				

Table S9. Calibrated Deferral Gate Regimes (τ_{cal}=0.04 Dev-Calibrated), First-Call Probe Escalation, Full OOD Baselines (N=6,108), & Continuous Streaming Tests. Under dual-arm vs. single-arm (final_only) feedback, the gate earns +8.69% net accuracy (+13.82%/$) in Compact-Primary cascades while acting as a neutral ablation on unconstrained 17-arm pools; first-call token probes raise accuracy to 82.25% (closing 27.7% oracle gap); and online bandit adaptation surpasses all offline, online neural, and greedy baselines under distribution shift.

Deferral Regime (Feedback)	Acc (%)	$/1k	Esc%	Net ΔAcc	ΔAcc/$
17-Pool Balanced (γ=0.08, None)	78.52	0.1578	0.0	0.00	—
17-Pool Balanced (both_arms / final_only)	78.45 / 78.55	0.1580	0.13	+0.02	+8.70
Probe-Augmented Router (Logprob+Ent)	82.25	0.1374	23.4	+3.66	+26.6
17-Pool Frugal (γ=0.30, None)	75.46	0.0669	0.0	0.00	—
Compact-Primary Tier (None, ≤ 8B)	69.27	0.0451	0.0	0.00	—
Compact-Primary (both_arms)	77.90	0.1482	41.91	+8.69	+13.82
Compact-Primary (final_only)	77.85	0.1481	41.01	+8.38	+13.40
OOD Shift 17-Pool (both_arms)	80.60	0.1282	0.23	+0.02	+2.48
Test-Time Hot Onboarding (Kimi-Think at t=500): First pick at 0 queries; 62.0% ([500, 550)) → 25.3% → 4.9% steady vs. 0% Frozen.					

OOD Benchmarks & Baselines	Method / Reg.	Acc (%)	$/1k
Static Frontier / Compact	Qwen-72B / Qianfan-8B	83.81 / 74.32	0.22 / 0.05
Offline Supervised	Frozen MLP / ARMS	56.97 / 66.58	0.07 / 0.11
Online SGD / Periodic Retrained	MLP+SGD / MLP (N=500)	71.45 / 76.80	0.12 / 0.14
Greedy Warm-Start (α=0)	Greedy Disjoint LinUCB	77.20	0.141
Neural Bandit / Lagrangian BwK	NeuralUCB / BwK-LinUCB	78.40 / 78.31	0.17 / 0.16
Proposed Online Bandits	CascadeRoute-VL / D-LinUCB	81.59 / 80.54	0.15 / 0.12
Gradual Drift / Repeated Switch	Frozen MLP / CascadeRoute	64.75 / 79.45	0.12 / 0.17
3× Price Shock (t>2k)	MLP (25.9% hi) / D-UCB (18.1%)	79.24 / 77.29	0.19 / 0.15
First-Call Probe Gains: +3.66% Acc, 27.7% Oracle Gap Closed (vs. FrugalGPT 76.45%, −4.7%).			

Table S10. Controlled Same-Backbone Study (Qwen3-VL-2B-Instruct vs. -Thinking), 12-Configuration Expansion, Single-Backbone Mode Router, Deconfounded Error Taxonomy, & Mechanism Probes (RTX 4050 4-bit CUDA, N=600 Runs). Complements Main Paper Fig. 7 and Table 4.

Same-Backbone Slice (N)	Inst Acc	Thk Acc	ΔAcc	OutTok (I/T)	p50 Lat (I/T)
OCRBench (10)	60.0%	70.0%	+10.0%	10.1 / 263.9	0.70s / 25.34s
ChartQA_TEST (10)	60.0%	40.0%	−20.0%	4.4 / 348.1	0.58s / 44.80s
MathVista_MINI (10)	10.0%	10.0%	0.0%	230.7 / 496.2	17.66s / 67.56s
HallusionBench (10)	40.0%	0.0%	−40.0%	2.0 / 349.1	0.34s / 49.92s
AMBER External (10)	90.0%	90.0%	0.0%	2.0 / 98.2	0.35s / 7.40s
ALL Stratified (50)	52.0%	42.0%	−10.0%	49.8 / 311.1	0.43s / 43.07s
Per-Encoder + LinUCB Latency (CUDA, N=50, p50/p95 ms): MiniLM-L6: 13.87/18.54;					
DINOv2: 15.20/18.86; ResNet-50: 16.50/19.81; optional 8 cues: 4.37/10.29; LinUCB: 1.01/1.45.					
Total 896-d / 904-d Overhead: 52.66 / 68.21 ms (904-d: 55.18ms p50).					

12-Config / External / Hardware Metric	Arms / N	Metric 1	Metric 2
1. Mode Only (HighRes, B512)	2 arms	Peak 52.0% @ 6.673s	Lat-AUC 45.38%
2. + Token Budget ({32, 128, 512})	6 arms	Peak 56.0%, 2.353s	Lat-AUC 55.64%
3. + Resolution (196^2, 392^2)	12 arms	Min Lat 1.327s	Lat-AUC 55.66%
CascadeRoute Mode Router (B512)	2 arms	54.0% Acc (50.0% OGC)	139.6 tok / 18.2s lat
Same-Backbone Mode Oracle	2 arms	56.0% Acc (100% OGC)	62.3 tok / 8.4s lat
External AMBER: CascadeRoute	10 ext	90.0% @ 0.350s	Oracle 100% @ 0.33s
Qwen3-VL-2B-Instruct OLS Calib.	300	p50/p95: 0.49s / 19.6s	MAE 0.598s, r=0.9894
Qwen3-VL-2B-Thinking OLS Calib.	300	p50/p95: 13.2s / 66.6s	MAE 0.661s, r=0.9985
Multi-Model Latency Calibration	6 models	$r \geq 0.9894$, MAPE 3.5%–14.0%	Full Table in Log

Deconfounded Failure Taxonomy (Qwen3-VL-2B HighRes B512): HallusionBench: 0.0% Thk Acc = 40.0% Trunc + 60.0% Genuine Error (relies on parametric priors); ChartQA: 40.0% Acc = 40.0% Trunc + 20.0% Genuine; OCRBench: 70.0% Thk vs 60.0% Inst = 30.0% Trunc + 0.0% Genuine Error. Mechanism Probe (Linguistic Hallucination Drift): Blank-Image Test drops Instruct −90% vs Thinking only −30%; Perturbation Flip Rate is 10.0% (Inst) vs 35.0% (Thk, p=0.008); Output Length vs Accuracy is r= − 0.086 (OCR) vs r= + 0.281 (STEM).

Table S11. Counterfactual OPE Audit Across 10 Loggers (N_{Test}=6,108, Uniformized ϵ=0.75 Logger). Previously, under severely imbalanced behavior propensities (Qwen2.5-VL-72B star-logger mismatch), SNIPS/DR confidence intervals undercovered true 72B accuracy; balanced propensity support yields 10/10 DR 95% CI coverage on both Qwen2.5-VL-72B and our Online Adaptive Trajectory (which adapts at runtime rather than a frozen snapshot).

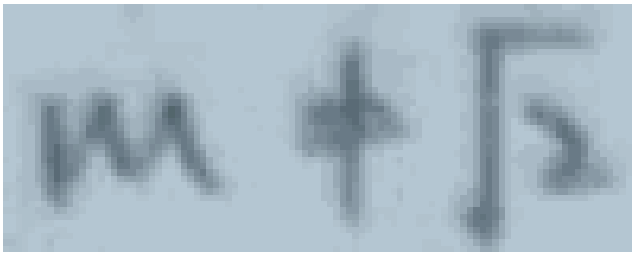
Evaluated Policy	True Acc	IPS (Seed 0 CI)	SNIPS (Seed 0 CI)	DR (Seed 0 CI)	Mean DR (10 Log)	10-Log Cov	Mean ESS
Always-DS-Tiny (Static)	0.6257	0.6414 [0.558, 0.740]	0.6017 [0.550, 0.658]	0.6140 [0.562, 0.670]	0.6274	9/10	300.3
Always-Qianfan-8B (Static)	0.6988	0.7086 [0.604, 0.809]	0.7222 [0.673, 0.771]	0.7282 [0.684, 0.776]	0.6984	10/10	299.4
Always-Qwen-72B (Static)	0.7729	0.7554 [0.663, 0.847]	0.7772 [0.728, 0.824]	0.7835 [0.739, 0.825]	0.7827	10/10	301.0
Offline-Linear	0.7883	0.7648 [0.731, 0.798]	0.7663 [0.749, 0.785]	0.7673 [0.747, 0.790]	0.7883	10/10	1789.9
MLP-Router (Static)	0.7988	0.8058 [0.723, 0.893]	0.8293 [0.794, 0.864]	0.8309 [0.797, 0.864]	0.8012	10/10	383.7
CascadeRoute (Warm-Start Snapshot)	0.7878	0.7873 [0.701, 0.871]	0.7925 [0.752, 0.825]	0.7942 [0.759, 0.829]	0.7883	10/10	418.0
CascadeRoute (Online Adaptive Trajectory)	0.7845	0.7921 [0.713, 0.879]	0.7862 [0.748, 0.826]	0.7894 [0.750, 0.826]	0.7808	10/10	397.7

Table S12. Domain Breakdown & 17-Arm Selection Distribution (Test N=6,108, Zero Task Labels). Unlisted compact/legacy arms (GPT-4o, Janus-1B/7B, LLaVA-7B, Pixtral-12B) receive 0.0% selection.

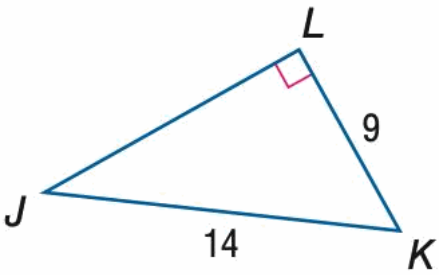
Domain (Test N)	72B Acc	Kimi Acc	Qianfan	Ours Acc	Ours OutTok	Active Arm	Overall (%)	STEM (%)	OCR (%)
STEM (1,631)	67.08	70.39	54.75	74.62	259.6	Qwen2.5-VL-72B	19.52	6.99	26.09
General_VQA (892)	73.21	69.84	66.48	73.77	102.7	Kimi-3B-Think	19.47	36.79	8.47
Charts_OCR (3,400)	83.56	75.00	78.29	81.91	4.9	Qianfan-VL-8B	17.31	4.66	26.03
Hallucination (185)	71.89	69.73	64.86	72.43	90.1	GeminiFlash2.5	11.10	10.97	12.06
						DeepSeek-VL2 (27B)	9.87	2.21	14.62
						MiMo-VL-7B-RL	8.97	26.36	0.09
						Qwen2.5-VL-32B	5.62	6.01	5.35
						InternVL2.5-78B	4.86	4.90	2.59
						DeepSeek-Tiny / Smol2	2.84	0.06	4.59



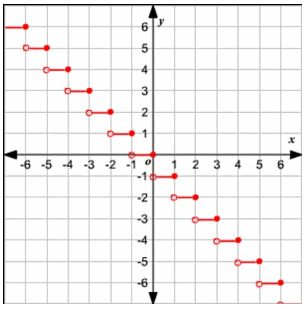
(a) OCRBench #314 (GT: Y809) — Qianfan-8B: Correct (7 tok); 72B/Think: Wrong.



(b) OCRBench #979 (GT: $m \neq \sqrt{2}$) — Qianfan-8B: Correct (8 tok); 72B/Think: Wrong.



(c) MathVista #897 (GT: 40) — Kimi-Think: Correct (178 tok); Direct/72B: Wrong.



(d) MathVista #874 (GT: 2) — Kimi-Think: Correct (269 tok); Direct/72B: Wrong.

Figure S1. Authentic VL-RouterBench Qualitative Gallery. CascadeRoute-VL learns to pair compact perception models with high-frequency scene text/formulas while routing multi-step STEM diagrams to reasoning specialists.

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[2] Miroslav Dudík, John Langford, and Lihong Li. Doubly robust policy evaluation and learning. In ICML, 2011.

[3] Adith Swaminathan and Thorsten Joachims. The self-normalized estimator for counterfactual learning. In NeurIPS, 2015.